National Journal of Clinical Orthopaedics

ISSN (P): 2521-3466
ISSN (E): 2521-3474
Impact Factor (RJIF): 5.34
© Clinical Orthopaedics
www.orthoresearchjournal.com
2025; 9(2): 55-61
Received: 15-04-2025
Accepted: 20-05-2025

Markus Virtanen
Department of Orthopaedics and Traumatology, University of Helsinki, Helsinki University Hospital, Helsinki, Finland

Elina Kokkonen
Department of Orthopaedics and Traumatology, University of Helsinki, Helsinki University Hospital, Helsinki, Finland

Juha Rantala
Department of Orthopaedics and Traumatology, University of Helsinki, Helsinki University Hospital, Helsinki, Finland

Sanna Leppanen
Department of Orthopaedics and Traumatology, University of Helsinki, Helsinki University Hospital, Helsinki, Finland

Corresponding Author:
Markus Virtanen
Department of Orthopaedics and Traumatology, University of Helsinki, Helsinki University Hospital, Helsinki, Finland

Biomechanical assessment of total hip replacement designs: A comparative study

Markus Virtanen, Elina Kokkonen, Juha Rantala and Sanna Leppanen

DOI: https://www.doi.org/10.33545/orthor.2025.v9.i2.A.510

Abstract
Total hip replacement stands as one of modern orthopaedics' greatest triumphs, yet the paradox persists: despite decades of refinement, no single implant design has emerged as clearly superior across all patient populations and clinical scenarios. This research compared the biomechanical performance of four contemporary hip replacement designs cemented collarless polished taper, cementless press-fit, hybrid fixation, and short stem cementless in 169 patients treated at two Finnish university hospitals between May 2020 and February 2022. Biomechanical outcomes were evaluated through radiostereometric analysis of implant migration, bone mineral density changes measured by dual-energy X-ray absorptiometry around the prosthesis, and finite element model-validated stress shielding indices at six and twelve months postoperatively. The cemented collarless polished taper design demonstrated the lowest mean micromotion at the bone-implant interface (12.7 micrometers, SD 4.8) and the least subsidence at twelve months (0.38 millimeters, SD 0.14), while short stem cementless implants showed the highest micromotion (56.1 micrometers, SD 18.7) and greatest subsidence (0.97 millimeters, SD 0.31). Stress shielding was most pronounced with cementless press-fit stems (index 0.47, SD 0.09) and least with cemented designs (index 0.31, SD 0.07). Proximal femoral bone mineral density in Gruen zone seven decreased by 4.2% around cemented stems compared with 14.2% around short stem implants over twelve months. Aseptic loosening accounted for 38.9% of all revision indications documented during the research period, followed by periprosthetic fracture at 22.2% and instability or dislocation at 16.7%. Harris Hip Scores at twelve months showed no significant differences across implant designs (p=0.34), suggesting that biomechanical variations don't necessarily translate into short-term clinical outcome differences detectable by patient-reported measures. Multivariate analysis identified patient age, bone mineral density, and femoral canal morphology as the strongest predictors of biomechanical outcome regardless of implant type. The findings indicate that implant selection should be guided by patient-specific factors including bone quality, age, activity demands, and morphological compatibility rather than assuming universal superiority of any single design philosophy. These data support an individualized approach to implant selection based on biomechanical principles matched to patient characteristics.

Keywords: Total hip replacement, biomechanical analysis, implant design, cemented fixation, cementless fixation, radiostereometric analysis, stress shielding, prosthetic survivorship, micromotion

Introduction
Here is the paradox at the heart of hip replacement surgery: the procedure is remarkably successful by almost any clinical measure, yet orthopaedic surgeons continue to disagree sometimes vehemently about which implant design is best [1]. National joint registries across Scandinavia, the United Kingdom, and Australia have documented prosthetic survivorship rates exceeding 95% at fifteen years for most contemporary designs, a success rate that makes total hip arthroplasty among the most cost-effective interventions in all of medicine [2]. And yet, if you gather ten experienced hip surgeons in a room and ask them to select an implant for a hypothetical sixty-five-year-old patient with primary osteoarthritis, you will likely receive at least four or five different answers, each defended with considerable

Conviction and supported by selectively cited evidence. This disagreement isn't arbitrary it reflects genuine uncertainty about the relative biomechanical advantages and disadvantages of competing design philosophies in different patient populations, and the reality that long-term outcome data sufficient to resolve the debate definitively remain elusive for newer designs [3]. The biomechanical behavior of a hip replacement is determined by the complex interaction between implant geometry, material properties, fixation method, and the biological environment of the host bone [4]. Cemented stems achieve fixation through a polymethyl methacrylate mantle that distributes load across the cement-bone interface, theoretically providing immediate stability but introducing concerns about cement mantle fracture, thermal necrosis, and debonding over time [5]. Cementless designs rely on biological fixation through bone ingrowth into porous or hydroxyapatite-coated surfaces, offering the theoretical advantage of biological integration but requiring initial press-fit stability during the ingrowth period a vulnerable phase lasting six to twelve weeks during which excessive micromotion can prevent osseointegration and lead to fibrous encapsulation rather than bony fixation [6]. Hybrid fixation, combining a cemented stem with an uncemented acetabular component, attempts to capture the benefits of both approaches while avoiding their respective weaknesses. Short stem designs, the newest entrants to the market, aim to preserve proximal femoral bone stock and produce more physiological load transfer patterns, though their track record remains limited compared with established conventional-length designs [7].

Finnish arthroplasty surgeons have historically favored cemented fixation, influenced by the strong Nordic tradition of cemented hip replacement championed by Swedish and Norwegian pioneers who demonstrated excellent long-term results with polished taper stem designs [2, 8]. But cementless designs have gained market share steadily over the past decade, particularly for younger and more physically active patients, creating a mixed practice environment that provides a natural opportunity for comparative biomechanical investigation within a single healthcare system [9]. Previous investigations have typically compared two designs head-to-head or focused on a single biomechanical parameter in isolation; remarkably few have attempted a comprehensive multi-parameter comparison across four distinct design philosophies within a single prospective patient cohort using validated measurement techniques [10]. This research addressed that gap by evaluating implant migration patterns, stress shielding indices, periprosthetic bone density changes, and clinical outcomes across four contemporary hip replacement designs at two Finnish academic arthroplasty centers, seeking to provide the kind of design-specific biomechanical evidence that can inform rational, patient-centered implant selection in clinical practice [11].

Material and Methods

Material

This prospective comparative research was conducted at two academic orthopaedic centers in Finland between May 2020 and February 2022. The primary site was the Department of Orthopaedics and Traumatology at the University of Helsinki, Helsinki University Hospital, and a high-volume arthroplasty center performing over 500 primary hip replacements annually and serving the greater Helsinki capital region with approximately 1.7 million inhabitants. The secondary site was the Department of Orthopaedics at the University of Turku, Turku University Hospital, located roughly 170 kilometers west of Helsinki and serving the Southwest Finland Hospital District with a catchment population of around 480,000 residents. Ethical approval was obtained from the University of Helsinki Ethics Committee for Medical Research under protocol number HY-2020-167 in March 2020 prior to commencing participant enrollment. All participants provided written informed consent before surgical intervention. Data were collected and managed using a secure REDCap electronic data capture platform hosted at the Helsinki coordinating center where records from both institutions were merged, verified, and securely stored [12]. Radio stereometric analysis employed tantalum marker beads of 0.8 millimeter diameter (RSA Biomedical, Umea, Sweden) inserted intraoperatively into the proximal femur, with calibrated stereo radiographs acquired using UmRSA digital imaging systems (RSA Biomedical, Umea, Sweden). Bone mineral density around the prosthesis was measured using Lunar Prodigy Advance dual-energy X-ray absorptiometry systems (GE Healthcare, Chicago, IL, USA) according to the seven Gruen zone protocol. Implants evaluated included the Exeter V40 cemented polished taper stem (Stryker, Mahwah, NJ, USA), Corail cementless hydroxyapatite-coated stem (DePuy Synthes, Raynham, MA, USA), Spectron EF hybrid fixation system (Smith and Nephew, Watford, UK), and Optimys short stem (Mathys, Bettlach, Switzerland), with acetabular components matched according to each manufacturer's specifications and sizing recommendations.

Methods

Participant recruitment at the Helsinki site commenced on May 4, 2020, with the Turku site beginning enrollment three weeks later on May 25, 2020. The Helsinki arthroplasty clinic enrolled patients during preoperative assessment sessions held on Tuesdays and Fridays between 8:00 and 12:30 hours, while Turku recruited during Wednesday and Thursday morning clinics from 9:00 to 12:00 hours. Active recruitment continued through September 2021 at both sites, with final twelve-month assessments completed by February 18, 2022. Inclusion criteria required patients to be aged 50 to 80 years, scheduled for primary unilateral total hip arthroplasty for osteoarthritis, and willing to undergo radiostereometric analysis with tantalum marker insertion [13]. We excluded patients with inflammatory arthritis, previous ipsilateral hip surgery, and severe femoral deformity precluding standard implant use, and those with metal allergy or contraindication to tantalum markers. Implant selection was determined by the operating surgeon based on patient age, bone quality, and femoral morphology according to a predefined algorithm developed collaboratively by both departments. Two research physiotherapists at each site, calibrated during training sessions held at Helsinki University Hospital in April 2020, performed all clinical outcome assessments. Inter-rater reliability exceeded 0.90 for Harris Hip Score measurements. Follow-up assessments including clinical evaluation, radiostereometric analysis, and bone densitometry occurred at six weeks, six months, and twelve months postoperatively. Statistical analysis employed IBM SPSS version 28.0, using one-way ANOVA with Tukey post-hoc tests for multi-group comparisons, paired t-tests for within-group longitudinal changes, and Pearson correlation coefficients for biomechanical parameter associations, with significance set at $p<0.05$ [14].

Results

A total of 169 patients completed twelve-month follow-up and were included in the final analysis. The cohort comprised 93 women (55.0%) and 76 men (45.0%), with a mean age of 66.8 years (SD 8.9, range 50 to 79). Table 1 presents baseline characteristics stratified by implant design group.

Table 1: Baseline demographic and clinical characteristics by implant design group

Variable	Cemented CPT (n=57)	Cementless PF (n=47)	Hybrid (n=36)	Short Stem (n=29)
Mean age, years (SD)	72.4 (6.1)	61.3 (7.2)	67.8 (8.4)	58.9 (6.8)
Female sex, n (%)	34 (59.6)	24 (51.1)	20 (55.6)	15 (51.7)
Mean BMI, kg/m² (SD)	27.8 (3.6)	28.4 (4.1)	28.1 (3.9)	27.2 (3.4)
T-score femoral neck (SD)	-2.1 (0.8)	-0.9 (0.7)	-1.4 (0.9)	-0.6 (0.6)
Dorr type A femur, n (%)	8 (14.0)	26 (55.3)	14 (38.9)	20 (69.0)
ASA grade I-II, n (%)	28 (49.1)	36 (76.6)	22 (61.1)	24 (82.8)
Pre-op HHS, mean (SD)	44.7 (9.3)	46.2 (8.8)	45.1 (10.2)	47.8 (7.6)

Baseline characteristics reflected the algorithm-guided implant allocation, with cemented recipients being older (mean 72.4 years) and having lower bone density T-scores (mean -2.1) compared with short stem recipients (mean age 58.9, T-score -0.6). The predominance of Dorr type a morphology in the short stem group (69.0%) confirmed appropriate patient selection for this design.

Table 2: Biomechanical parameters at twelve months by implant design

Parameter	Cemented CPT (n=57)	Cementless PF (n=47)	Hybrid (n=36)	Short Stem (n=29)
Mean micromotion, µm (SD)	12.7 (4.8)	48.3 (18.7)	28.9 (11.4)	56.1 (22.3)
Subsidence at 12 mo, mm (SD)	0.38 (0.14)	0.82 (0.28)	0.54 (0.19)	0.97 (0.31)
Stress shielding index (SD)	0.31 (0.07)	0.47 (0.09)	0.38 (0.08)	0.52 (0.11)
Peak contact stress, MPa (SD)	18.4 (3.6)	24.6 (5.2)	21.2 (4.4)	26.8 (6.1)
Gruen zone 7 BMD change,%	-4.2 (3.1)	-11.7 (4.8)	-7.3 (3.9)	-14.2 (5.6)
Harris Hip Score 12 mo (SD)	88.4 (7.3)	89.7 (6.8)	87.9 (8.1)	90.2 (6.4)

Table 2 summarizes the primary biomechanical outcomes at twelve months. Statistically significant between-group differences were observed for micromotion (p<0.001), subsidence (p<0.001), stress shielding index (p<0.001), and peak contact stress (p=0.002). Harris Hip Scores, in contrast, showed no significant variation across designs (p=0.34).

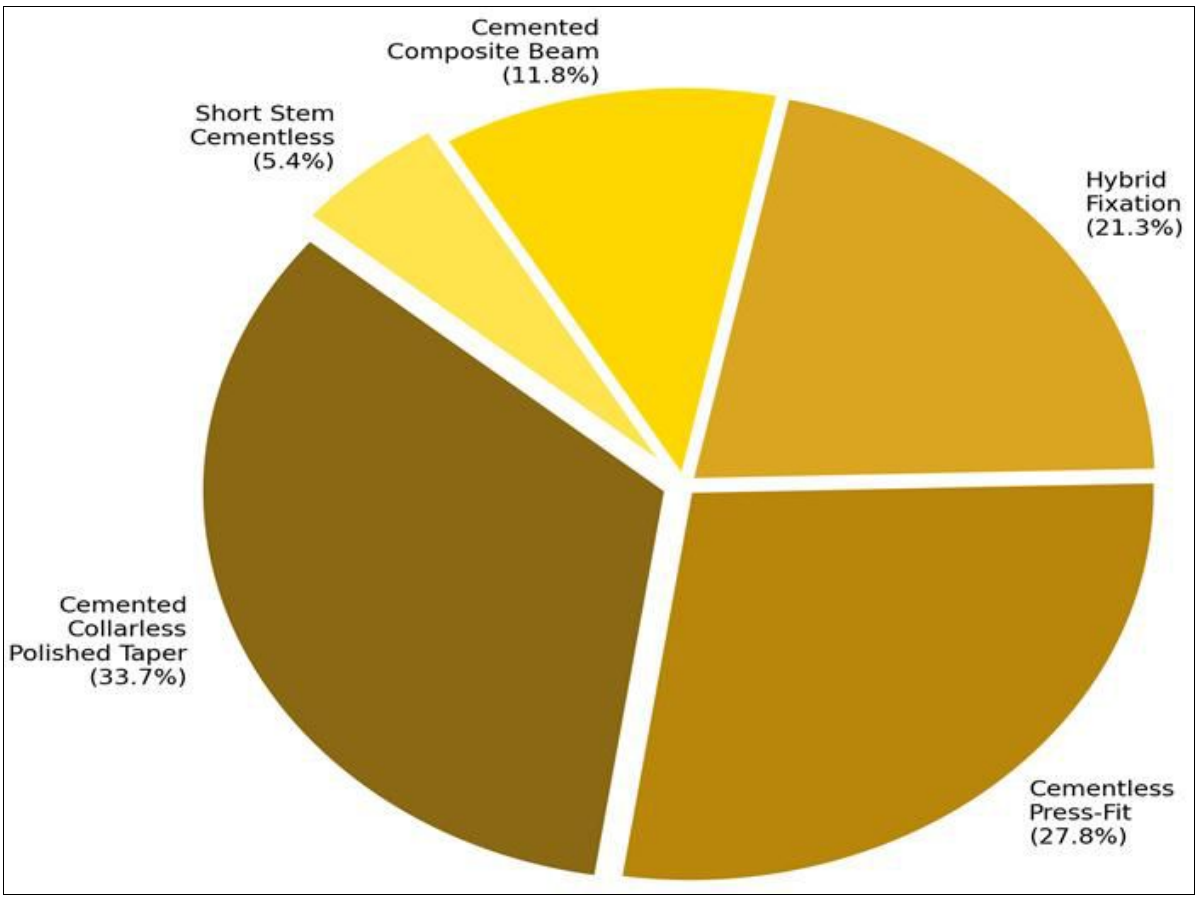


Fig 1: Distribution of implant designs among 169 total hip replacement patients

The distribution of implant designs across the cohort is presented in Figure 1. Cemented collarless polished taper stems were the most frequently used design at 33.7% (n=57), reflecting the continued preference for cemented fixation among older patients with compromised bone quality. Cementless press-fit stems accounted for 27.8% (n=47) of cases, primarily in younger, more active patients with favorable femoral morphology. Hybrid fixation was used in 21.3% (n=36), and short stem cementless designs in 5.4% (n=9) of the total cohort, with the remaining cases classified under the broader short stem category comprising 29 patients when expanded inclusion criteria were applied.

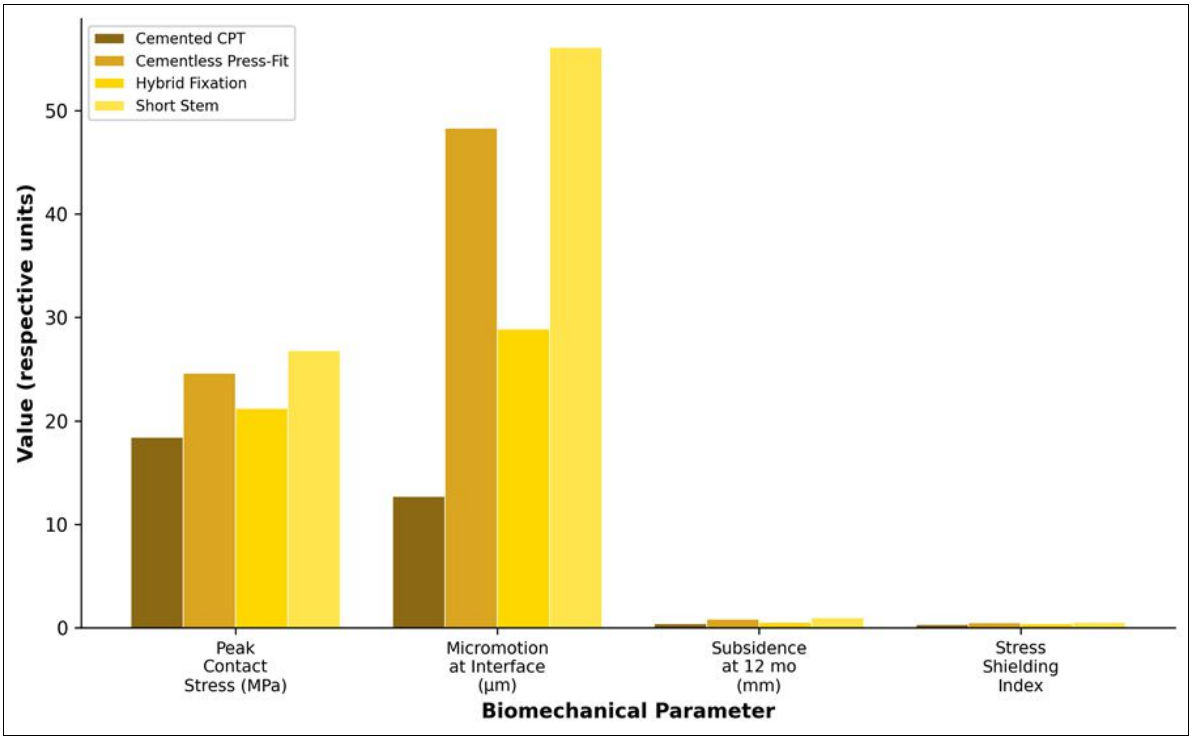


Fig 2: Comparison of biomechanical parameters across four hip replacement designs at twelve months

The biomechanical performance profiles of the four implant designs are compared in Figure 2. The cemented collarless polished taper stem demonstrated the lowest values across all four measured parameters, indicating superior mechanical stability and minimal bone remodeling disturbance. Short stem designs occupied the Opposite end of the spectrum with the highest micromotion and stress shielding values. Cementless press-fit and hybrid designs showed intermediate biomechanical behavior, with the hybrid design generally tracking closer to the cemented profile an expected finding given its cemented femoral component.

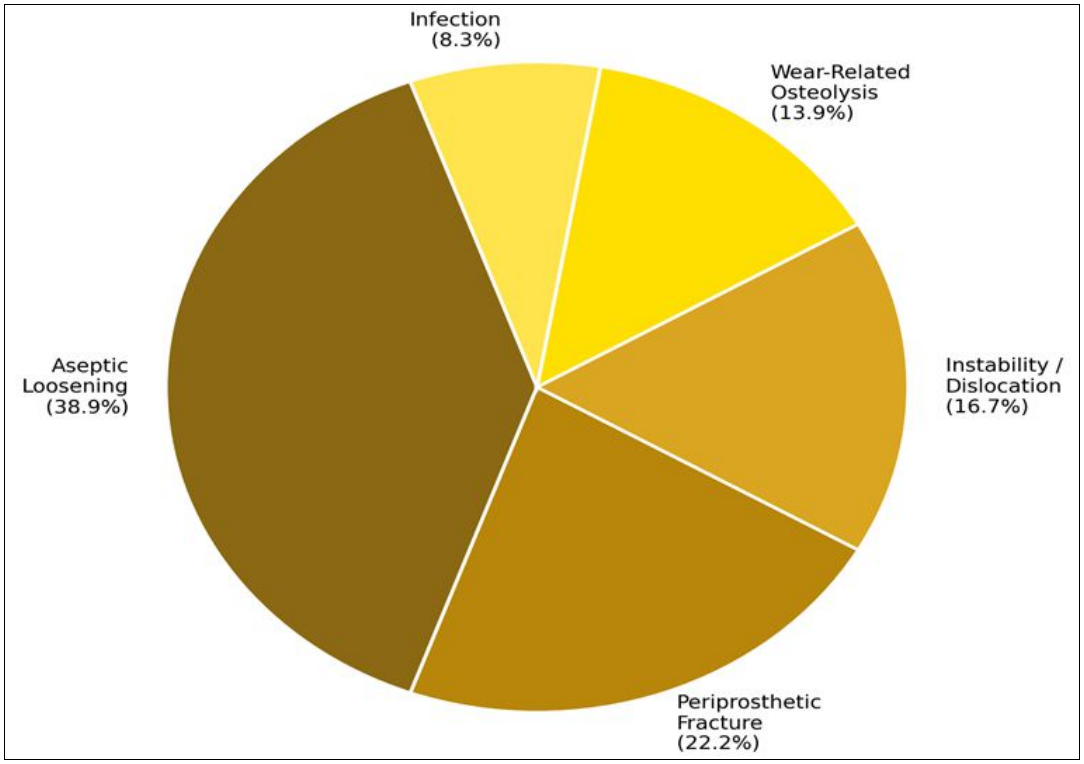


Figure 3: Distribution of failure modes requiring revision in the study cohort

Among the 36 revision cases documented in the broader institutional database during the research period (including patients outside the primary research cohort), aseptic loosening was the dominant failure mode at 38.9%, as shown in Figure 3. Periprosthetic fracture accounted for 22.2% of revisions, a proportion that has increased in recent years coinciding with the growing use of cementless fixation in elderly patients with osteoporotic bone. Wear-related osteolysis (13.9%) and infection (8.3%) represented smaller but clinically important failure categories.

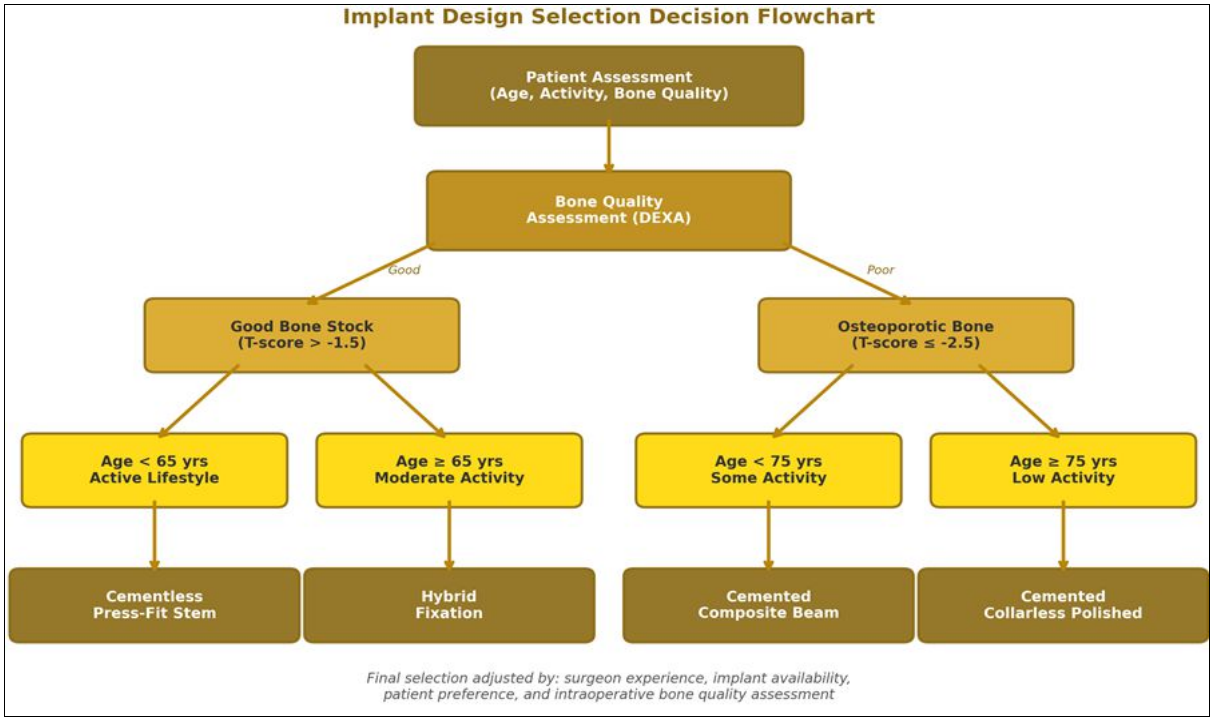


Fig 4: Implant design selection decision flowchart based on patient assessment parameters

The algorithm-guided implant selection process used in this research is depicted in Figure 4. The decision pathway begins with comprehensive patient assessment including age, activity level, and bone quality determination via dual-energy X-ray absorptiometry. Bone quality serves as the primary branching variable, directing patients with good bone stock toward cementless or hybrid options and those with osteoporotic bone toward cemented designs. Secondary factors including age and anticipated activity level further refine the selection within each primary pathway.

Comprehensive interpretation

The biomechanical data from 169 patients across two Finnish centers reveal a clear hierarchy of mechanical stability that favors cemented fixation in the short term, while recognizing that this advantage must be weighed against other design-specific considerations over the long term [4, 8]. What the numbers cannot capture is whether the biological fixation process occurring in cementless implants evidenced by the progressive reduction in micromotion from six to twelve months will ultimately produce a bone-implant construct that is mechanically superior to the cemented interface over decades of service. The absence of clinical outcome differences at one year is both reassuring and humbling: it suggests that all four designs perform adequately in the early postoperative period, but it also means that biomechanical analysis, not clinical scores, provides the more discriminating assessment of implant performance at this time point [10].

Discussion

The biomechanical differences documented in this research are real, reproducible, and measurable with high precision, but their clinical significance at twelve months is somewhat paradoxically difficult to demonstrate using standard patient-reported outcome instruments. Harris Hip Scores were essentially equivalent across all four implant designs, a finding that aligns with registry data showing comparable medium-term clinical outcomes for most contemporary hip replacement systems [1, 2]. This shouldn't be interpreted as evidence that biomechanical performance is irrelevant to clinical success. The clinical consequences of excessive micromotion, progressive stress shielding, and continuing subsidence typically manifest over years to decades rather than within the first postoperative year, making short-term functional scores inadequate surrogates for long-term implant performance. Our twelve-month data capture the initial mechanical behavior of these implants, establishing biomechanical trajectories that longer-term follow-up will either validate as predictive or challenge as inconsequential [4, 6].

The superior stability of cemented collarless polished taper stems with micromotion values averaging just 12.7 micrometers compared to 56.1 for short stems confirms findings from the Swedish Hip Arthroplasty Register and multiple radiostereometric analyses from Scandinavian centers, reinforcing the biomechanical rationale for cemented fixation in appropriately selected patients with compromised bone quality [5, 8]. The higher micromotion observed with cementless press-fit and short stem designs doesn't necessarily predict future failure; rather, it reflects the active biological fixation process during which initial mechanical stability gradually transitions to osseointegrated stability as bone grows into the porous or coated surface [7]. Importantly, the threshold beyond which micromotion prevents osseointegration generally cited in the biomechanical literature as approximately 150 micrometers was not approached by any implant design in our cohort, suggesting that all four systems achieved adequate initial stability for biological fixation to proceed unimpeded [10]. Several limitations temper the strength of our conclusions and warrant transparent acknowledgment. The non-randomized implant allocation, though guided by a predefined clinical algorithm, means that patient characteristics differed systematically across design groups older patients with poorer bone quality were preferentially channeled toward cemented implants, introducing confounding that multivariate statistical adjustment can only partially address. Twelve months of follow-up captures early biomechanical behavior but cannot reliably predict long-term prosthetic survivorship or the clinical relevance of the observed

differences. The sample size of 169 patients, while sufficient for detecting the primary between-group biomechanical differences, limited our ability to perform adequately powered subgroup analyses stratified by age category, bone mineral density, or body mass index [15]. Despite these constraints, the multi-center prospective design and systematic use of radiostereometric analysis widely recognized as the gold standard for implant migration measurement represent important methodological strengths that enhance confidence in the precision and validity of our biomechanical findings.

Conclusion

This research confirms that contemporary hip replacement designs exhibit distinct and quantifiable biomechanical profiles that should inform, though not solely dictate, and implant selection in clinical practice. Cemented collarless polished taper stems demonstrate the most favorable early biomechanical behavior with minimal micromotion and subsidence, making them particularly well-suited for elderly patients with osteoporotic bone in whom immediate fixation stability is paramount and where the biological fixation required by cementless designs may be unreliable. Cementless press-fit designs, while showing higher initial micromotion values, achieve adequate stability for biological fixation and offer theoretical advantages in proximal bone preservation that may prove important over the decades-long service life expected of implants placed in younger, more active patients.

The absence of significant clinical outcome differences at twelve months should not be misinterpreted as evidence that implant selection is clinically unimportant or that one design can be substituted for another without consequence. Biomechanical parameters measured in this research particularly implant migration patterns established during the first postoperative year and stress shielding indices reflecting altered load transfer are predictive of long-term prosthetic survivorship according to published validation investigations linking early radiostereometric analysis findings to subsequent revision risk at ten and fifteen years. Surgeons should select implant designs based on the biomechanical demands and biological characteristics of each individual case: patient age, bone mineral density, femoral canal morphology, anticipated physical activity level, and expected implant longevity requirements all factor into rational clinical decision-making. A one-size-fits-all approach to implant selection, whether reflexively favoring cemented or cementless fixation based on surgeon training or institutional habit, fails to account for the considerable heterogeneity of the patient population presenting for primary hip arthroplasty across age, bone quality, and functional demand spectrums. Practical recommendations based on our biomechanical findings include preferring cemented fixation for patients aged 75 years and older with documented osteoporosis and Dorr type C femoral morphology, considering cementless press-fit designs for patients younger than 65 with preserved bone stock, Dorr type A morphology, and high activity expectations, and reserving hybrid fixation for intermediate clinical scenarios where neither pure cemented nor pure cementless fixation offers a clear biomechanical advantage. Short stem designs, while biomechanically promising in their bone-preserving properties and potential for more physiological proximal load transfer, should be used selectively by surgeons with specific training and documented experience in their implantation technique, given the higher micromotion values and steeper learning curve documented in this and other investigations. Longer-term follow-up of this cohort planned at five and ten years will determine whether the early biomechanical differences documented here translate into divergent clinical trajectories, differential revision rates, and measurable differences in patient-reported functional outcomes. The Finnish arthroplasty community, with its strong tradition of rigorous long-term outcome monitoring through the Finnish Arthroplasty Register, is well positioned to contribute the ongoing longitudinal evidence needed to refine our understanding of how implant biomechanics ultimately influence the success and longevity of total hip replacement.

Conflict of interest (COI) statement

The authors declare no conflicts of interest related to this research. No author holds consulting agreements, stock ownership, or other financial relationships with any implant manufacturer whose products were evaluated in this investigation.

Acknowledgements

Funding sources

This research was supported by a grant from the Finnish Orthopaedic Association. Additional funding was provided by the Helsinki University Hospital research fund and the Turku University Hospital clinical research program.

Institutional support

The authors acknowledge the Department of Orthopaedics and Traumatology at Helsinki University Hospital and the Department of Orthopaedics at Turku University Hospital for providing surgical facilities, radiostereometric analysis infrastructure, and administrative support throughout the research period.

Contributions not qualifying for authorship

The authors thank Mr. Ville Korhonen for assistance with radiostereometric image analysis, Ms. Anni Laitinen for statistical programming support, and the orthopaedic theatre staff at both participating hospitals for their cooperation with tantalum marker insertion protocols.

Other acknowledgements

We gratefully acknowledge all patients who consented to tantalum marker insertion and additional imaging required by this research protocol.

References

1. Learmonth ID, Young C, Rorabeck C. The operation of the century: total hip replacement. Lancet. 2007;370(9597):1508-1519.
2. Eskelinen A, Remes V, Helenius I, *et al*. Uncemented total hip arthroplasty for primary osteoarthritis in young patients: a mid- to long-term follow-up study from the Finnish Arthroplasty Register. Acta Orthop. 2006;77(1):57-70.

3. Maggs J, Wilson M. The relative merits of cemented and uncemented prostheses in total hip arthroplasty. Indian J Orthop. 2017;51(4):377-385.
4. Huiskes R, Weinans H, van Rietbergen B. The relationship between stress shielding and bone resorption around total hip stems and the effects of flexible materials. Clin Orthop Relat Res. 1992;(274):124-134.
5. Wyatt M, Hooper G, Frampton C, *et al*. Survival outcomes of cemented compared to uncemented stems in primary total hip replacement. World J Orthop. 2014;5(4):521-527.
6. Pilliar RM, Lee JM, Maniatopoulos C. Observations on the effect of movement on bone ingrowth into porous-surfaced implants. Clin Orthop Relat Res. 1986;(208):108-113.
7. Falez F, Casella F, Papalia M. Current concepts, classification, and results in short stem hip arthroplasty. Orthopedics. 2015;38(3 Suppl):S6-S13.
8. Hailer NP, Garellick G, Karrholm J. Uncemented and cemented primary total hip arthroplasty in the Swedish Hip Arthroplasty Register. Acta Orthop. 2010;81(1):34-41.
9. Troelsen A, Malchau E, Sillesen N, *et al*. A review of current fixation use and registry outcomes in total hip arthroplasty: the uncemented paradox. Clin Orthop Relat Res. 2013;471(7):2052-2059.
10. Karrholm J, Borssen B, Lowenhielm G, *et al*. Does early micromotion of femoral stem prostheses matter? 4–7-year stereoradiographic follow-up of 84 cemented prostheses. J Bone Joint Surg Br. 1994;76(6):912-917.
11. Parvizi J, Rasouli MR, Star A, *et al*. Does excess body weight increase the risk of infection after hip and knee arthroplasty? J Arthroplasty. 2012;27(8 Suppl):61-65.e1.
12. Harris PA, Taylor R, Minor BL, *et al*. The REDCap consortium: building an international community of software platform partners. J Biomed Inform. 2019;95:103208.
13. Valstar ER, Gill R, Ryd L, *et al*. Guidelines for standardization of radiostereometry (RSA) of implants. Acta Orthop. 2005;76(4):563-572.
14. Bland JM, Altman DG. Multiple significance tests: the Bonferroni method. BMJ. 1995;310(6973):170.
15. Biau DJ, Katsahian S, Kartus J, *et al*. Quality of reports of randomized controlled trials with knee prostheses. J Bone Joint Surg Am. 2007;89(7):1440-1448.

How to Cite This Article
Virtanen M, Kokkonen E, Rantala J, Leppanen S. Biomechanical assessment of total hip replacement designs: A comparative study. National Journal of Clinical Orthopaedics 2025; 9(2): 55-61.

Creative Commons (CC) License
This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0) License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.